

Report: MAM4 Project

Project Study Report

Testing Climate Strategies through Simulation

MAM 4
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Polytech Nice Sophia

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1 Introduction and Objectives

Climate change is now one of the major challenges of the 21st century. The increase in greenhouse gas concentrations, primarily linked to human activities, is leading to a gradual rise in the global average temperature and profoundly altering environmental, economic, and social balances. In this context, the Paris Agreement sets the goal of limiting global warming to well below 2 °C above pre-industrial levels, and pursuing efforts to limit it to 1.5 °C. Achieving these goals requires a profound transformation of energy, industrial, and land-use systems on a global scale.

Analyzing these transformations requires a systemic approach capable of representing the complex interactions between energy, economy, land, and climate. Integrated assessment models, based on system dynamics (stocks, flows, and feedback loops), make it possible to study these interactions and evaluate the impact of different technological and policy choices on emissions and temperature over the long term.

As part of this project, we used the interactive climate simulator *En-ROADS* (Energy Rapid Overview and Decision-Support), developed by Climate Interactive in collaboration with the MIT Sloan School of Management and Ventana Systems. This tool allows testing different climate policy levers (carbon price, energy efficiency, electrification, renewable energy development, land management, etc.) and observing their effects on global emissions, the energy mix, and temperature evolution up to 2100. Designed to run quickly in a web browser while being calibrated based on scientific literature, it is a particularly suitable tool for learning about and analyzing climate strategies.

The objective of this project is to verify whether the different trajectories associated with the commitments of the Paris Agreement are realistic, using a system dynamics model. This involves understanding the main internal mechanisms of the model, particularly its structure in terms of stocks, flows, and feedback, then linking climate policy levers to the variables that influence emissions and temperature. We also seek to analyze the links between policy decisions (levers) and climate outcomes, to compare several transition scenarios, and to discuss the coherence and limits of the obtained trajectories. Through this approach, this work mobilizes the tools of applied mathematics and modeling to provide a quantitative analysis of strategies to combat climate change.

2 Work Plan and Logbook

Here is how our project study unfolded throughout the MAM4 semester:

2.1 October 2025

- Topic assignment
- Meeting with Ms. Chlela, our tutor throughout this project
- Familiarization with the En-ROADS climate simulator, on which we will base all our simulations

2.2 November 2025

- More detailed exploration of the En-ROADS model
- Creation of the first graphs
- Selection of key variables
- Project follow-up meeting with Ms. Chlela to define a plan to follow, as well as the key concepts to address/encounter throughout the project

2.3 December 2025

- Highlighting and study of the sub-systems related to our key variables
- Selection and study of the most effective levers to activate
- Creation of several scenarios
- Graph extraction
- Project follow-up meeting with Ms. Chlela

2.4 January 2026

- Review of the current situation
- In-depth analysis of each sub-system
- Representation of the main causal chains
- Project follow-up meeting with Ms. Chlela

2.5 February 2026

- In-depth analysis of the scenarios
- Creation of a comprehensive scenario
- Highlighting the model's limitations
- Project follow-up meeting with Ms. Chlela
- Beginning of report writing

2.6 March 2026

- End-of-project follow-up meeting with Ms. Chlela
- Finalization of the report writing
- Oral presentation

3 The En-ROADS Model

3.1 How It Works

First, it is essential to present the main tool we used throughout this project to establish our hypotheses and forecasts: the En-ROADS climate simulator.

The En-ROADS model is a global climate simulation model based on system dynamics. It aims to represent in an integrated manner the interactions between three main components: energy systems (production, consumption, energy mix, electrification), land and agricultural systems (land use, deforestation, carbon sequestration), and the climate system (greenhouse gas emissions, atmospheric concentration, and global temperature evolution).

The model adopts a systemic approach: these different components are interconnected and their evolutions influence each other.

Stocks and Flows Logic: En-ROADS is based on a stocks and flows logic. Stocks represent quantities accumulated over time, such as the atmospheric CO₂ concentration, forest areas, or installed capacities of energy production. Flows correspond to the variations that modify these stocks, for example, annual CO₂ emissions, deforestation, or the installation of new renewable capacities.

The Differential Equations at the Heart of the Model From a mathematical perspective, En-ROADS is based on a system of differential equations that govern the evolution of each stock over time. For a given stock, its variation is expressed according to the fundamental equation of system dynamics:

$$\frac{dS}{dt} = \sum(\text{inflows}) - \sum(\text{outflows})$$

where S represents the considered stock, and where inflows and outflows themselves depend on other variables of the system, thus creating a network of coupled equations.

Unlike an optimization approach that would seek a single optimal solution, En-ROADS uses these equations to perform a simulation: starting from historical data (since 1990), the model calculates step by step the evolution of the system by projecting the interactions between the different variables.

Numerical Resolution and Choice of Time Step These differential equations being non-linear and coupled, they cannot be solved exactly analytically. The model therefore proceeds to an iterative numerical resolution, using the Euler method (or derived methods).

To ensure the accuracy of the results while maintaining a reasonable computation time, the model works with a fixed time step of 0.125 years (about 45 days). This time step is deliberately small for several reasons:

- It allows for capturing finely the variations of flows and their feedbacks
- It reduces the numerical error accumulated at each iteration
- It ensures that the simulated solution remains as close as possible to the exact solution of the equation system

Over the entire simulation (from 1990 to 2100, i.e., 110 years), this small time step allows for several thousand calculations, thus ensuring a fine modeling of energy-climate dynamics.

Feedback Loops: The model integrates positive and negative feedback loops, which mathematically translate into couplings between equations. Positive loops amplify a phenomenon, for example when the increase in temperature accentuates ice melt and reinforces warming. Negative loops, on the contrary, tend to stabilize the system, such as when an increase in the carbon price reduces fossil fuel consumption and therefore emissions.

These mechanisms reflect the complex interactions between the different variables of the system and are fully captured by the system of differential equations.

Representation of Technologies and Policies: En-ROADS allows acting on several levers representing technological and policy choices: development of renewable energies, improvement of energy efficiency, electrification of uses, energy storage, reforestation, or implementation of a carbon tax. Each lever corresponds to a parameter of the model. When the user modifies a lever, the equations are recalculated with these new values and the effects propagate throughout the system through feedbacks.

Simulator Functioning: The simulator runs in a web browser and allows for "what-if" analyses. The user can adjust the various levers and instantly observe the consequences of these modifications up to 2100. The results are presented in the form of graphs showing the evolution of emissions, CO₂, the energy mix, and global temperature.

Link with Mathematical Modeling: It is therefore through this system of coupled differential equations, solved numerically with a finite time step, that En-ROADS models energy gains, the carbon cycle, and climate feedbacks. Using this model allowed us to mobilize our skills in modeling, dynamic systems, and quantitative analysis to study a concrete and global problem: evaluating climate transition strategies.

3.2 Key Variables

Once we mastered the En-ROADS simulator, our objective was to identify the most relevant variables to select for our future modeling. We therefore varied a large number of parameters to observe their respective effects on emissions, CO₂ concentration, and temperature by 2100. This exploratory phase allowed us to distinguish variables with a significant impact from those whose effect remained marginal or highly dependent on other mechanisms.

Table 1 below summarizes the variables we identified as central to our analysis. For each one, we specified the concerned sub-system (energy, demand, land, or climate), their role in the model's dynamics (stock, flow, or auxiliary variable), their unit, as well as the expected effect when activating a typical lever, such as increasing a carbon tax.

Table 1: Key variables identified via the En-ROADS simulator

Variable	Sub-system	Role	Unit	Expected effect if carbon tax ↑
Relative price of fossil fuels	Energy (prices)	Aux.	\$/MWh	Increase
Share of solar/wind	Energy (prices)	Aux.	%	Increase
Energy emissions	Climate / Emissions	Flow	GtCO ₂ /yr	Decrease
CO ₂ Concentration	Climate	Stock	ppm	Decrease (with delay)
Installed fossil capacity (power plants)	Energy / Infrastructure	Stock	GW	Decrease
Fossil energy production	Energy	Flow	EJ/yr or TWh/yr	Decrease
Carbon intensity of energy	Energy / Performance	Aux.	tCO ₂ /EJ or gCO ₂ /kWh	Decrease
Final energy demand	Demand / Uses	Flow	EJ/yr or TWh/yr	Decrease
Electrification rate	Technologies / Demand	Aux.	%	Increase
Deforestation	Land	Flow	Gha/yr	Decrease (slight)
Agricultural yield	Land	Aux.	t/ha or % yield	Constant (tax doesn't act directly; impact via temperature if climate change)
Global mean temperature	Climate	Stock	°C	Decrease (with delay)
Total GHG emissions	Climate	Flow	GtCO ₂ /yr	Decrease
Natural CO ₂ absorption	Climate	Flow	GtCO ₂ /yr	Slight increase

This classification helped us better understand the internal structure of the model and identify the most strategic entry points. We thus observed that some variables play a structuring role in the long term, notably stocks (installed capacities, atmospheric concentration), while others act more directly on annual emission flows. This distinction guided the construction of our scenarios and the analysis of the causal chains presented below.

4 Sub-systems

4.1 Selection of Sub-systems

Once the initial table of key variables was established, we chose to deepen the analysis by focusing on the sub-systems to which these variables belong. This step allowed us to structure the model into coherent sets and facilitate its interpretation.

We thus distinguished four main sub-systems: the Energy sub-system, the Energy Demand and Uses sub-system, the Land and Land Use sub-system, and finally the Climate and Carbon Cycle sub-system.

To continue our study, we first conducted a phase of detailed understanding of each of these sub-systems taken in isolation. The objective was to identify their internal mechanisms, their dominant stocks and

flows, as well as their main causal chains. This individual analysis then allowed us to compare their respective impacts and better understand their role in the overall dynamics of the model.

4.2 Reading the Diagrams

4.2.1 Energy Sub-system

Stocks: Fossil capacities and renewable capacities constitute stocks, as they represent infrastructure accumulated over time. They introduce strong inertia into the energy system: new low-carbon capacities are added gradually, while existing installations continue to produce until the end of their lifetime. This inertia reflects a temporal and non-linear dimension of the system. The energy transition is therefore necessarily progressive, highlighting the importance of acting early.

Flows: Investments modify installed capacities. Energy production and energy emissions are annual flows that reflect the system's activity in each period.

Auxiliary Variables: The carbon tax, relative energy prices, the share of solar and wind, and the carbon intensity of energy are auxiliary variables. They do not accumulate but influence flows via causal relationships.

Key Message: Economic signals guide investment flows, which progressively modify the stock of capacities and ultimately determine the emission level of the energy sector.

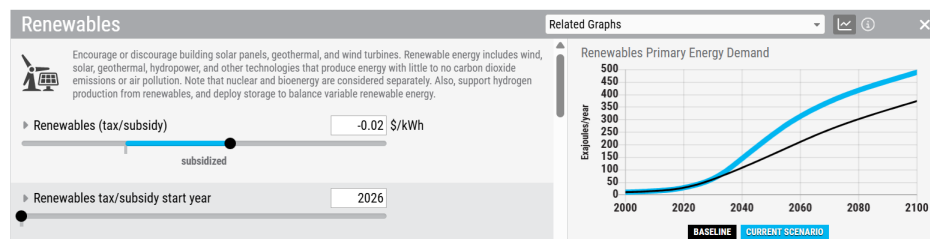


Figure 1: Renewable energies

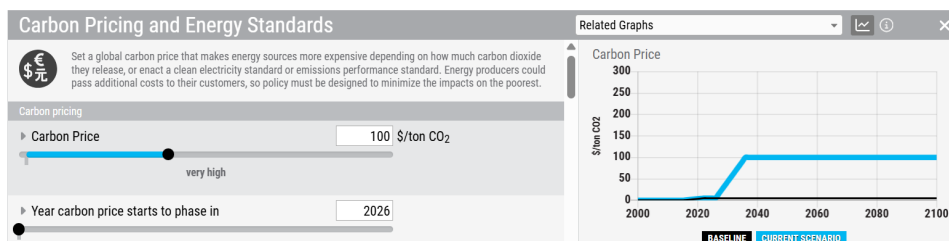


Figure 2: Carbon tax

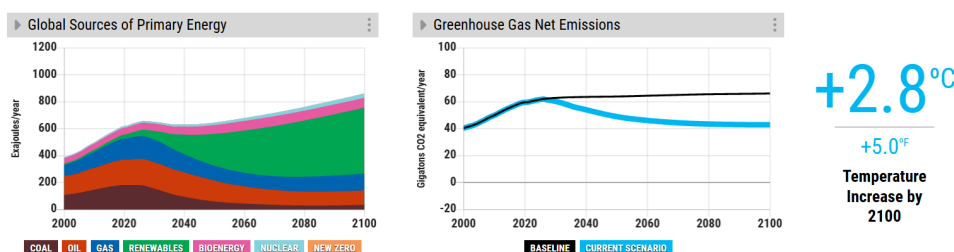


Figure 3: Carbon tax + Renewable energies

4.2.2 Energy Demand and Uses Sub-system

Reducing energy demand and transforming uses is a central lever for the transition, as it acts directly on energy and emission flows.

Stocks: There is no major physical stock explicitly represented in the simplified diagram. However, equipment (vehicles, buildings, industrial machinery) constitutes implicit stocks that introduce inertia: their renewal is slow, creating a technological lock-in phenomenon.

Flows: Final energy demand is an annual flow (EJ/yr or TWh/yr). It depends on the level of economic activity, energy efficiency, and uses. Total energy production adjusts to meet this demand. A decrease in demand mechanically leads to a reduction in associated emissions.

Auxiliary Variables: Energy efficiency, the electrification rate, and the carbon intensity of energy modulate the flows:

- An improvement in energy efficiency reduces the amount of energy needed for the same service.
- Electrification modifies the structure of uses.
- Carbon intensity determines emissions per unit of energy consumed.

Main Causal Chains:

Efficiency $\uparrow \rightarrow$ Demand $\downarrow \rightarrow$ Production $\downarrow \rightarrow$ Emissions $\downarrow$

Electrification $\uparrow \rightarrow$ Carbon intensity $\downarrow \rightarrow$ Emissions $\downarrow$

Key Message: The transition does not rely solely on decarbonizing the supply side. Acting on the amount of energy consumed and on the nature of uses makes it possible to directly reduce emission flows, often more quickly than transforming the production system alone.

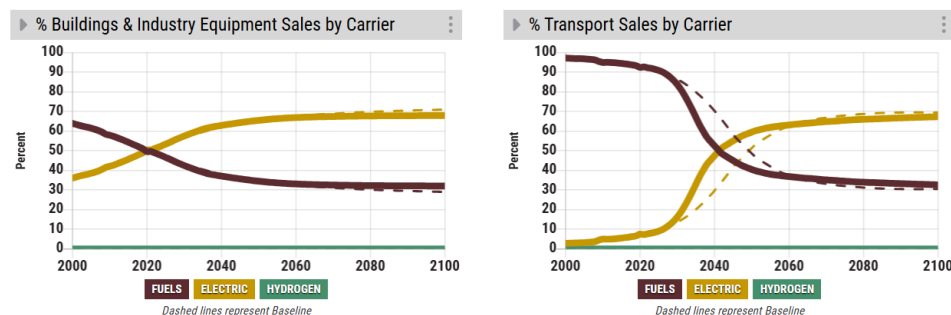


Figure 4: Electrification of buildings and transport

The graphs illustrate the energy transition in two sectors. In buildings and industry, the shift from fuels to electricity is gradual. In transport, the shift is abrupt and concentrated around 2035/2040, with a near disappearance of fossil fuels in favor of electricity.

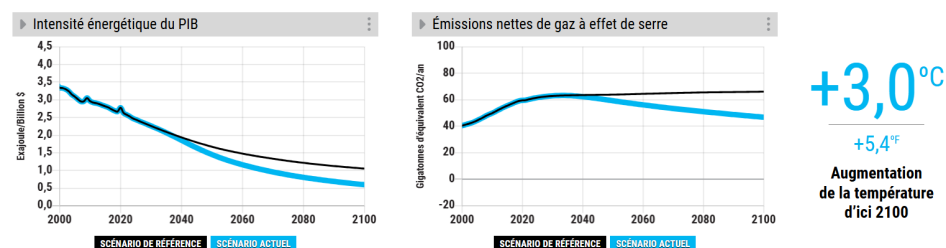


Figure 5: Electrification and energy efficiency of buildings and transport

4.2.3 Land and Land Use Sub-system

Stocks: The main stock is terrestrial carbon (forests, soils, biomass). It constitutes a natural reservoir that evolves slowly, introducing ecological inertia.

Flows: Deforestation decreases the terrestrial carbon stock. Afforestation and reforestation increase it. These flows determine the net emissions or absorptions related to land use.

Auxiliary Variables: Agricultural yield indirectly influences pressure on land. Higher yields can limit the opening of new agricultural areas. Forest policies also modulate the flows.

Main Causal Chains:

Deforestation $\uparrow \rightarrow$ Carbon stock $\downarrow \rightarrow$ CO₂ $\uparrow$

Afforestation $\uparrow \rightarrow$ Carbon stock $\uparrow \rightarrow$ CO₂ $\downarrow$

Stabilizing Loop:

Forest stock $\uparrow \rightarrow$ Absorption $\uparrow \rightarrow$ CO₂ $\downarrow$

Key Message: The land system acts as a carbon sink or source depending on the balance between deforestation and reforestation. It is essential for achieving "Net Zero," but remains limited by biophysical constraints.

4.2.4 Climate and Carbon Cycle Sub-system

Stocks: The main stock is the atmospheric CO₂ concentration (ppm), the accumulation of past emissions. The global mean temperature can also be considered a stock due to the thermal inertia of the oceans.

Flows: Total GHG emissions constitute an inflow into the atmosphere, while natural absorption (oceans, forests) constitutes an outflow. Their difference determines the evolution of the CO₂ stock.

Auxiliary Variables: Radiative forcing links GHG concentration to the temperature increase by translating the Earth's energy imbalance.

Main Causal Chains:

Emissions $\uparrow \rightarrow$ CO₂ $\uparrow \rightarrow$ Forcing $\uparrow \rightarrow$ Temperature $\uparrow$

Positive Climate Loop:

Temperature $\uparrow \rightarrow$ Absorption $\downarrow \rightarrow$ CO₂ $\uparrow \rightarrow$ Temperature $\uparrow$

Key Message: The climate operates with strong inertia and feedbacks. Current emissions accumulate in an atmospheric stock that affects temperature with a delay. Even a rapid reduction in emissions only causes a gradual slowdown in warming, hence the importance of acting early.

4.3 Analysis of Sub-systems

4.3.1 Energy Sub-system

Where does it act? : This sub-system acts on the energy supply and the carbon intensity of the system.

Main Mechanism:

- investments $\rightarrow$ capacities $\rightarrow$ production $\rightarrow$ emissions
- We gradually replace fossil capacities with low-carbon capacities.

Type of Impact:

- structural and lasting emission reduction.
- progressive effect (infrastructure inertia).
- major long-term impact.

Temporality:

- slow initially (inertia of existing plants).
- very powerful in the long term.

Limitation:

- if demand remains high, a huge amount of low-carbon capacity is needed. The transition is therefore more expensive and slower.

4.3.2 Demand & Use Sub-system

Where does it act? : This sub-system acts on the total quantity of energy consumed.

Main Mechanism:

- efficiency → lower demand.
- electrification → lower carbon intensity (if the mix is clean).

Type of Impact:

- direct and immediate reduction of energy flows.
- can produce rapid effects.
- reduces pressure on the energy system.

Temporality:

- faster than the energy sub-system.
- depends on equipment renewal.

Limitation:

- if electricity remains carbon-intensive, electrification alone is insufficient.

4.3.3 Land & Land Use Sub-system

Where does it act? : This sub-system acts on natural carbon sinks.

Main Mechanism:

- increased afforestation → increased carbon stock → increased absorption.
- decreased deforestation → fewer emissions.

Type of Impact:

- acts on outflows (absorption).
- can offset part of the emissions.
- important for Net Zero.

Temporality:

- slow (forest growth).
- progressive effect over several decades.

Limitation:

- limited by available land area.
- cannot absorb high fossil emissions indefinitely.

4.3.4 Climate and Carbon Cycle Sub-system

Where does it act? : It is the final physical outcome of the system.

Main Mechanism:

- emissions - absorption $\rightarrow$ atmospheric CO₂ $\rightarrow$ temperature

Type of Impact:

- translates the effects of the other sub-systems.
- introduces significant delays.
- contains feedbacks (positive loops).

Temporality:

- very slow (inertia of CO₂ and oceans).

Limitation:

- we cannot act directly on temperature; only via emissions and absorption.

Table 2: Summary table of sub-systems and their characteristics

Sub-system	Acts on	Type of effect	Speed	Long-term power
Energy	Supply / Carbon intensity	structural	slow	very strong
Demand & Use	energy quantity	direct	medium/fast	strong
Land & Land Use	carbon absorption	compensatory	slow	limited but useful
Climate & Carbon Cycle	CO ₂ stock / temperature	final result	very slow	depends on others

5 Policy Parameters / Levers - Table 2**5.1 Methodology for Selecting Levers**

Based on our understanding of the sub-systems, we identified the most effective levers through an empirical sensitivity analysis on En-ROADS. By varying each lever independently, we observed their impact on the temperature in 2100 and the emissions trajectory. This allowed us to distinguish three categories of levers:

- **Structural levers:** They profoundly transform the energy system (major long-term impact).
- **Flow levers:** They act directly on demand and uses (faster effects).
- **Compensatory levers:** They increase carbon absorption (physically limited potential).

5.2 Parameterization of Levers for Scenarios

For each of the selected levers, we defined precise implementation hypotheses: value, start date, political constraints, and acceptability conditions. Table 3 summarizes these choices, which served as the basis for constructing our scenarios.

Table 3: Policy parameters of the levers (linked to hypotheses and scenarios)

Lever	Value(s)	Start	Political Constraint	Hypotheses (including acceptability)	Expected Result(s)
Carbon tax	100 \$/t	2028	Budget j= ...	Compensation for low-income households	Temperature increase between 2.7°C and 2.9°C by 2030 instead of 3.3°C
Final efficiency	+2.5%/yr	2026	Progressive standards	Renovation aid	Decrease in total consumption by energy source (temperature drop of 0.2°C in 2030)
Afforestation	58% of available land	2027	Limited land	Certification / monitoring	From 2085, +3 GtCO ₂ /yr absorbed
Transport Electrification	25% purchase cost subsidy	2026	Insufficient charging infrastructure	Rapid decrease in battery cost	Electric transport sales = fuel transport sales by 2039 (instead of 2046)
Buildings & industry electrification	25% purchase cost subsidy	2026	Slow fleet renewal + high investments	Gradual adoption due to falling costs	By 2040: 62.25% of new equipment electric (vs 59.88%); 37.75% fossil fuels (vs 40.12%)
Methane & other gases	-40% reduction	2026	Resistance from agricultural sector (livestock)	Change in agricultural practices / diets	Decrease in agricultural methane intensity
Deforestation	-50%/yr reduction	2027	Local economic pressure	International aid (payment for ecosystem services)	Net CO ₂ emissions from land use: 5.75 Gt/yr in 2060 (vs 6.71 Gt/yr without measure)
Renewables	0.02 \$/kWh subsidy	2027	Landscape acceptability	Progress in storage to manage intermittency	Significant increase in low-carbon energy consumption

5.3 Which Levers Have the Most Impact?

Table 4: Comprehensive comparison of levers: parameters, impact, durability, and feasibility

Lever	Value	Start	Durability	Impact T° 2100	Feasibility / Key Question
Carbon tax	100\$/t	2028	Very durable (30-40 years)	High (-0.5°C)	<i>Is it realistic?</i> Yes (Sweden, Canada), but social acceptability must be ensured
Energy efficiency	+2.5%/yr	2026	Durable (equipment renewal)	High (-0.4°C)	<i>Can we renovate massively?</i> Yes but slow and costly
Electrification	25% subsidy	2026	Very durable (equipment 10-20 years)	Very high (-0.7°C)	<i>Can we electrify that much?</i> Technically yes, but huge budget
Renewables	subsidy	2027	Very durable (infrastructure)	High (-0.5°C)	<i>Is it financeable?</i> Yes, but storage needs development
Forests	58% land	2027	Durable but limited (saturation)	Low (-0.1°C)	<i>Is it enough?</i> No, physical limit
Methane reduction	-40%	2026	Limited (short-lived gas)	Medium (-0.2°C)	<i>Is it possible?</i> Yes but agricultural barriers

This table shows that the carbon tax and electrification are the most durable levers because they profoundly transform infrastructure (supply and uses). Forests and methane are useful complements but insufficient on their own.

5.4 Study of Two Major Levers

5.4.1 The Carbon Tax: A Price Signal to Transform Supply

Mechanism:

Tax $\uparrow$ $\rightarrow$ Fossil prices $\uparrow$ $\rightarrow$ Low-carbon investments $\uparrow$ $\rightarrow$ Renewable capacities $\uparrow$ $\rightarrow$ Emissions $\downarrow$

Strengths: Powerful systemic lever in the long term, modifies the entire system via the price signal. Durable effect (low-carbon plants last 30-40 years).

Weaknesses: Depends on political acceptability, progressive effect (inertia of existing fleet). Requires compensation for low-income households.

Example: Sweden, carbon tax since 1991 (over €130/t) $\rightarrow$ emissions decreased while maintaining growth.

5.4.2 Electrification: A Transformation of Uses

Mechanism:

Electrification $\uparrow$ + Clean mix $\rightarrow$ Carbon intensity of uses $\downarrow \rightarrow$ Emissions $\downarrow$

Strengths: Transforms physical stocks (vehicles, boilers), cumulative and durable effect, acts directly on uses.

Weaknesses: Requires a decarbonized electricity mix and charging infrastructure. Depends on the rate of equipment renewal.

Examples:

- In Norway, over 80% of car sales are electric.
- In China, massive electrification + renewables.

5.5 Synthesis

Among all the levers, the carbon tax is the most durable because it acts on production infrastructure for 30-40 years. Electrification follows with an effect over 10-20 years (equipment renewal). Forests, although useful, have a limited effect because they saturate and act very slowly (after 2080). Thus, acting on supply (energy) and demand (electrification) is more structuring in the long term than natural sinks alone.

No single lever alone will be enough to meet the Paris Agreement goals. Indeed:

- **Carbon tax and renewables:** structurally decarbonize supply.
- **Efficiency and electrification:** reduce demand and transform uses.
- **Forests and methane reduction:** essential supporting roles (buy time, offset residual emissions).

We conclude that it is the simultaneous combination of these levers, creating a synergy effect, that will allow us to envisage a trajectory compatible with climate objectives.

6 Selected Scenarios

6.1 Current Scenario (Reference)

The reference scenario, or "Business as Usual" (BAU), corresponds to the continuation of current trends without implementing new ambitious climate policies. Implementing this in En-ROADS, this scenario would take into account the following parameters:

- Carbon tax: non-existent or very low
- Energy efficiency: slow progression (1-1.5%/yr)
- Electrification: current trends without acceleration
- Renewable energies: moderate development
- Deforestation: continues at the current rate
- Methane: no targeted reduction

Thus, the projected result in 2100 would be: an increase in global average temperature of 3.3°C compared to the pre-industrial era.

This scenario serves as a comparison point to evaluate the effect of the levers activated in our prospective scenarios.

6.2 Our Choices

To build our scenarios, we selected the most impactful levers identified previously. Table 5 presents our simulations, taking into account a gradual implementation of measures to reflect reality.

Table 5: Experiment log (log of simulations performed)

Test	Lever(s)	Value(s)	Start	Hypothesis	Recorded Indicators
1	CO ₂ Tax	100 \$/t	2028	Substitution towards renewables	Emissions, renewables share, temperature
2	Efficiency + Electrification	+2.5%/yr *	2026	First reduce energy quantity, then make it cleaner. Gradual implementation over 10 years to reach this rate.	Impact on consumption and energy bills
3	Methane + Other gases	-40%	2026	Rapid impact on temperature (short-lived gases)	60.52 GtCO ₂ /yr in 2050; reduction of 0.2°C by 2100
4	Nature (Forests)	+3 Gt (Affo) / -50% (Defo)	2027/2028	Creation of carbon sinks for "Net Zero"	CO ₂ absorption, land use; reduction of 0.1°C by 2100
5	Comprehensive scenario	All previous levers	2026-2030	Acting on multiple levers reinforces emission reduction	Temperature 2100: +2.3°C (vs +3.3°C without measures)

* For energy efficiency, the rate of +2.5%/yr was retained because it represents a significant effort (close to the maximum possible 3% in En-ROADS) while remaining realistic. In our simulations, this rate is achieved progressively over 10 years, with annual improvements, which better corresponds to the reality of climate policies.

6.3 Forecasts and Impacts of Each Scenario

6.3.1 Scenario 1 - Carbon Tax Alone

Sub-systems activated: Energy (mainly), Climate (indirectly)

Mechanism: The carbon tax increases the price of fossil fuels → decrease in fossil investments → relative increase in renewables → gradual decrease in energy emissions.

Observed effects:

- Decrease in energy sector emissions
- Increase in the share of renewables
- Moderate temperature decrease in the long term

Limitation: The action mainly affects energy supply. Energy demand remains relatively high → reductions are partial.

Conclusion: This is therefore an effective lever but insufficient alone to meet climate goals.

6.3.2 Scenario 2 - Energy Efficiency + Electrification

Sub-systems activated: Demand & Use, Energy, Climate

Mechanism:

- Efficiency → lower energy demand
- Electrification → reduction in the carbon intensity of uses (if the mix is low-carbon)

Observed effects:

- More direct reduction in total consumption
- Fewer emissions related to uses (transport, buildings, industry)
- Faster impact than the carbon tax alone

Limitation: Highly dependent on the energy mix. If electricity remains carbon-intensive, the gain is limited.

Conclusion: Acting on demand is a powerful lever because it immediately reduces energy and emission flows.

6.3.3 Scenario 3 - Methane and Other Gas Reduction

Sub-systems activated: Climate (mainly), Land & Agriculture (indirectly)

Mechanism: Reduction of short-lived gases → rapid decrease in radiative forcing.

Observed effects:

- Relatively rapid impact on temperature
- Visible effect before 2100
- 60.52 GtCO₂/yr in 2050 vs 58.25 GtCO₂/yr in 2100 (reduction of 0.2°C)

Limitation: Does not act on the accumulated CO₂ stock, so does not solve the long-term structural problem.

Conclusion: A useful complementary lever to slow warming in the short term.

6.3.4 Scenario 4 - Nature-Based Solutions

Sub-systems activated: Land & Land Use, Climate

Mechanism: Increase in terrestrial carbon stock → more CO₂ absorption.

Observed effects:

- Creation of a carbon sink
- Progressive reduction of atmospheric CO₂
- 63.67 GtCO₂/yr in 2050 vs 61.32 GtCO₂/yr in 2100 (reduction of 0.1°C)

Limitations:

- Slow effect (forest growth)
- Physical limits (land availability)

Conclusion: Important for achieving Net Zero, but cannot offset fossil emissions alone.

6.3.5 Scenario 5 - Comprehensive Scenario (Integrated Approach)

Sub-systems activated: Demand & Use, Energy, Climate, Land

Mechanism:

- Demand reduction (efficiency)
- Decarbonization of supply (carbon tax, renewables)
- Transformation of uses (electrification)
- Protection of natural sinks (forests)
- Reduction of other GHGs (methane)

Observed effects:

- Significant decrease in emissions
- Stronger temperature reduction by 2100: +2.3°C vs +3.3°C without measures
- Synergy effect: the levers reinforce each other

Conclusion: It is therefore the combination of levers that allows a trajectory compatible with limiting warming. No single sub-system alone will suffice.

7 Comprehensive Scenario

7.1 Differences between Isolated Scenarios and the Comprehensive Scenario

Unlike isolated scenarios that activate one or two levers, the comprehensive scenario combines all actions in a coordinated manner. The table below presents the fundamental differences between these two approaches.

Table 6: Key differences between isolated and integrated approaches

Aspect	Isolated lever approach	Integrated approach
Effect	Additive (sum of individual effects)	Multiplicative (synergies)
Temporal coverage	Short term OR long term	Short term AND long term
Resilience	Low (depends on a single lever)	High (redundancy and complementarity)
Achieving 2°C	Impossible	Possible

7.2 Synergies between Sub-systems

The strength of the comprehensive scenario lies not in adding up levers, but in their interactions. Each sub-system reinforces the effectiveness of the others, creating a multiplier effect.

- **Synergy 1: Demand ↔ Energy**

Energy efficiency reduces total demand, which decreases the volume of renewable capacity needed. The transition becomes less costly and faster.

- **Synergy 2: Electrification ↔ Renewables**

Electrification of uses increases electricity demand. Combined with renewable development (carbon tax + subsidies), it ensures that this new demand is met by low-carbon sources.

- **Synergy 3: Carbon tax ↔ Acceptability**

Energy efficiency gains reduce household bills, partially offsetting the impact of the carbon tax and improving its social acceptability.

- **Synergy 4: Temporality**

Methane reduction acts quickly (slowing warming by 2040), while energy system transformations bear fruit in the long term. Forests play a compensatory role for residual emissions after 2080.

7.3 Analysis of the Graphs

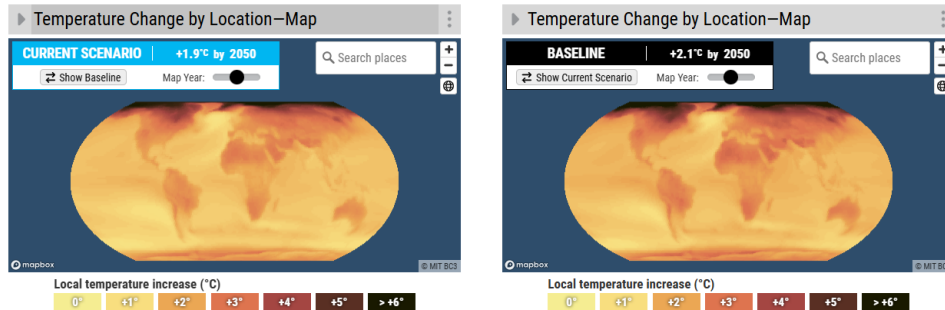


Figure 6: Map of temperature increase in 2050

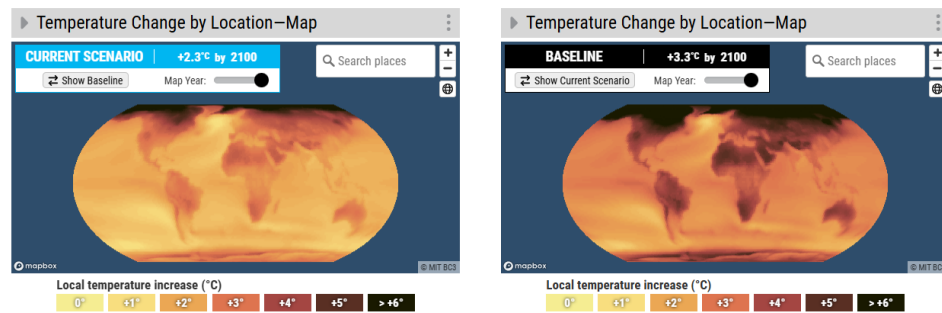


Figure 7: Map of temperature increase in 2100

These two images compare the increase in global temperature under two scenarios: on the left, the reference scenario; on the right, our comprehensive scenario. The first image corresponds to the year 2050, the second to 2100. It can be observed that in the absence of ambitious measures, the gap widens over time: in 2050, warming reaches +2.1 °C in the reference scenario, compared to +1.9 °C in our comprehensive scenario. By 2100, this gap widens considerably, with +3.3 °C for the reference scenario, compared to +2.3 °C for our scenario.

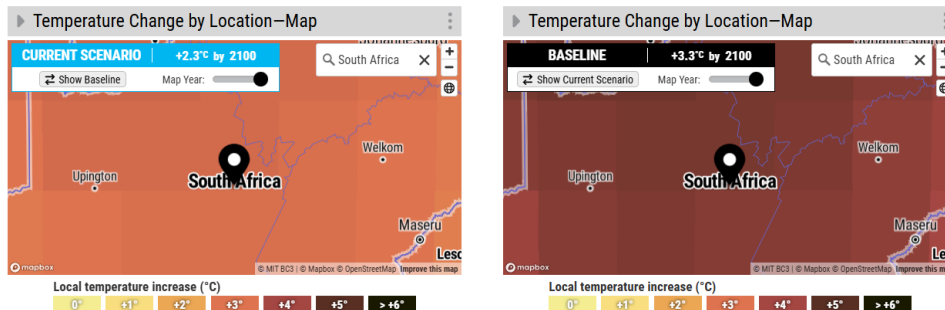


Figure 8: South Africa in 2100

Taking South Africa as an example, we observe a difference of more than one degree in temperature increase between the two scenarios by 2100.

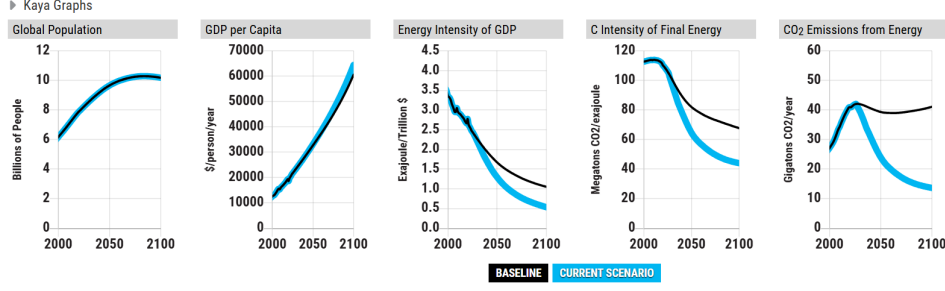


Figure 9: Kaya Graph

The Kaya graphs decompose CO₂ emissions according to the following identity:

$$CO_2 = P \times \frac{GDP}{P} \times \frac{E}{GDP} \times \frac{CO_2}{E}$$

where P is the world population, E is energy consumed, $\frac{E}{GDP}$ is the energy intensity of GDP, and $\frac{CO_2}{E}$ is the carbon intensity of energy.

Comparing the comprehensive scenario (blue curve) to the reference scenario (black curve), we observe that:

- Population and GDP per capita follow similar trajectories in both scenarios: economic and demographic growth continues until 2100.
- The energy intensity of GDP and the carbon intensity of energy drop much more sharply in the comprehensive scenario, reflecting significantly more ambitious energy efficiency gains and decarbonization of the energy mix.
- Consequently, CO₂ emissions peak around 2025 and then collapse to about 15 Gt in 2100 in the comprehensive scenario, compared to ~ 40 Gt in the reference scenario.

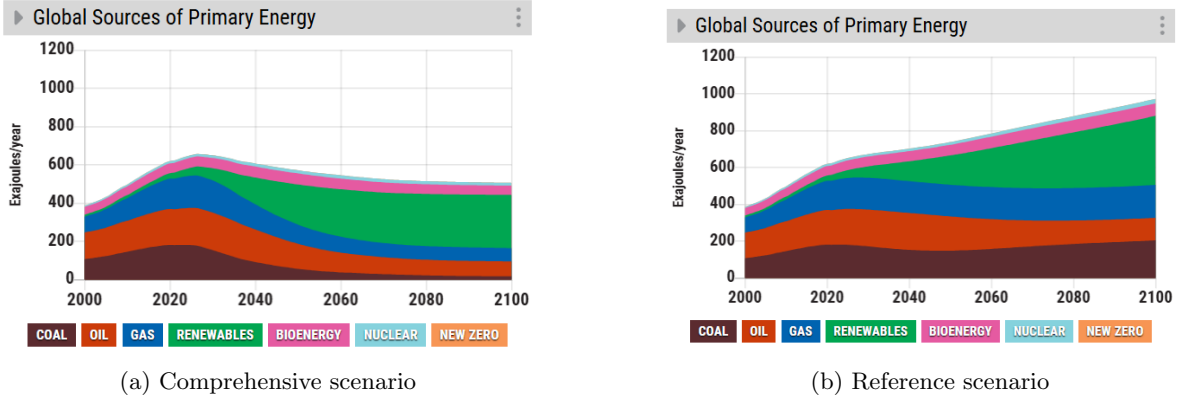


Figure 10: Global primary energy sources

While the **reference scenario** projects a continuous increase in primary energy consumption up to ~950 EJ/yr in 2100, dominated by fossils, the comprehensive scenario shows a peak around 2030 followed by a decrease to ~450 EJ/yr. This trajectory relies on the gradual phase-out of coal and oil in favor of renewable energies and new low-carbon technologies.

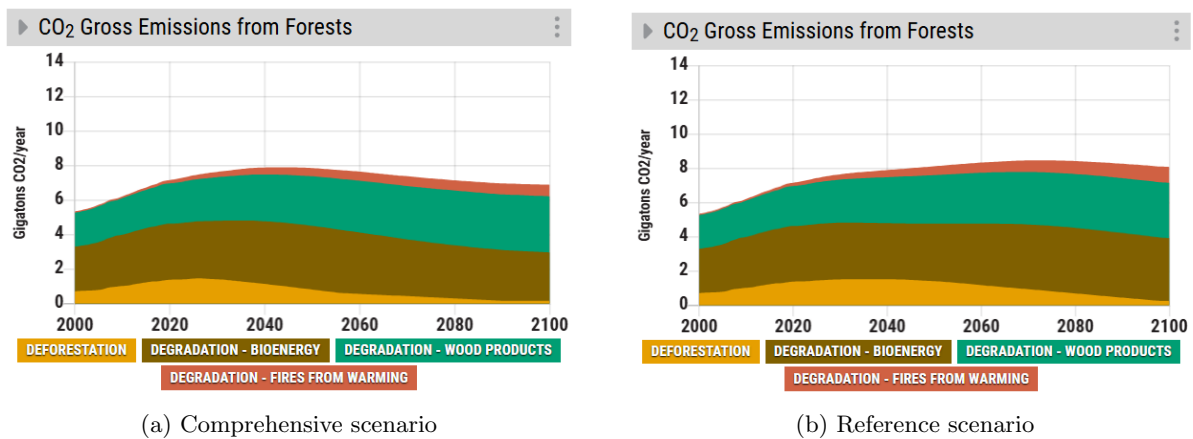


Figure 11: Gross CO₂ emissions from forests (Gt/yr)

Both graphs show gross CO₂ emissions from forests. The two scenarios show very similar trajectories, with a peak around 2040–2060 reaching about 8 Gt CO₂/yr, followed by a slight decline. The difference remains limited: the comprehensive scenario slightly reduces emissions at the end of the century (about 7 Gt vs 8 Gt in the reference scenario), mainly due to a reduction in deforestation.

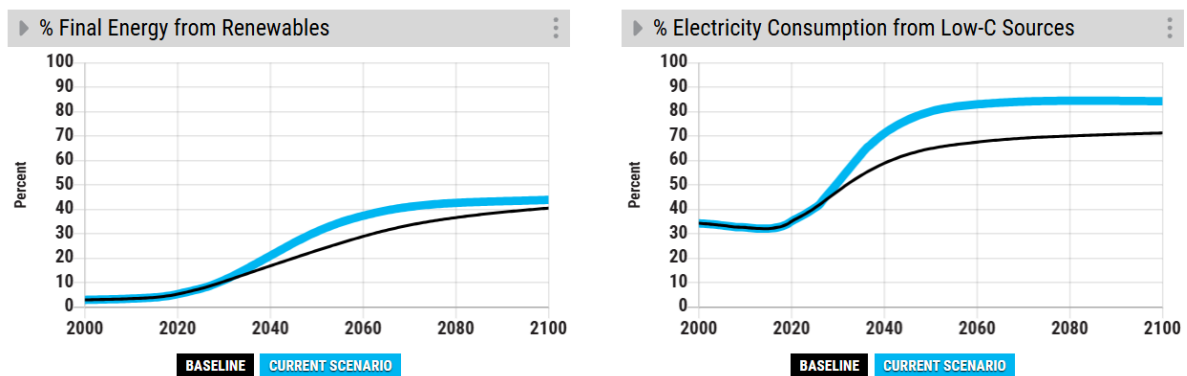


Figure 12: Share of renewables in final energy and share of low-carbon sources in electricity consumption (%)

Both graphs show an increase in low-carbon energy in both scenarios, but faster in the comprehensive scenario. The share of renewables in final energy reaches about 45 % in 2100 compared to nearly 40 % in the reference scenario. Similarly, the share of electricity from low-carbon sources increases more sharply in the comprehensive scenario, reaching about 85 % by the end of the century compared to about 70 % in the reference scenario.

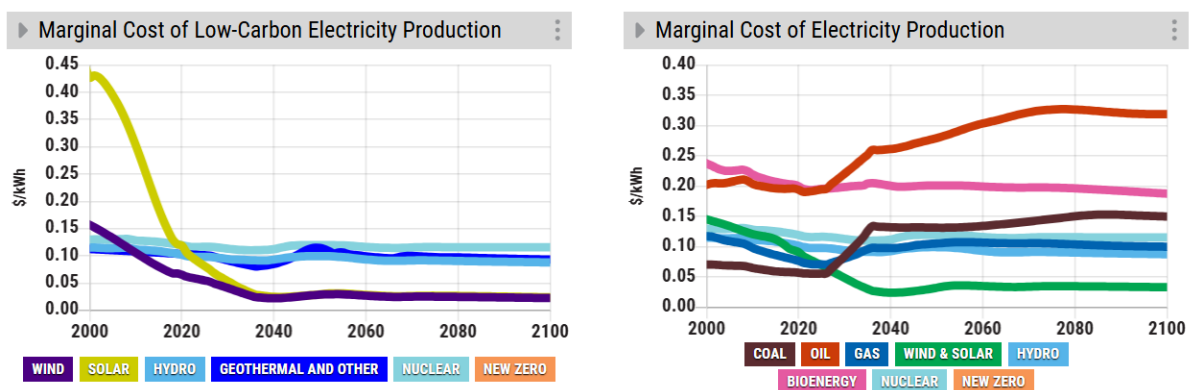


Figure 13: Marginal costs of electricity production

In our comprehensive scenario, solar power experiences the most dramatic cost reduction, falling from \$0.43/kWh in 2000 to less than \$0.03/kWh in 2100, while onshore wind converges to \$0.02/kWh. These two sources thus become the cheapest. Conversely, fossil fuels (oil, gas) see their marginal costs increase after 2030, widening the competitiveness gap in favor of renewables.

7.4 Expected Result

This comprehensive scenario would reach +2.3°C by 2100, i.e., a gain of -1°C compared to the scenario without action (+3.3°C), compatible with the Paris Agreement. This result proves that no single isolated lever is sufficient; only a combined approach would allow us to meet climate goals.

8 Parallel with Reality

8.1 Where Do We Stand Currently?

Ten years after the Paris Agreement, COP30, held in Belém (Brazil) in November 2025, delivered a mixed assessment. While climate multilateralism was maintained and notable progress was recorded – notably the commitment to mobilize 1.3 trillion dollars per year by 2035, the doubling of adaptation finance, and the operational activation of the “loss and damage” fund – the final agreement falls short of the necessary ambition.

No binding roadmap was adopted concerning the gradual phase-out of fossil fuels, the main driver of greenhouse gas emissions. Similarly, halting deforestation, crucial for preserving natural carbon sinks, was not subject to legally binding measures. However, the official recognition of the fight against climate disinformation is an important symbolic step to anchor action in science.

8.2 Are the Goals Achievable? Do We Need to Make More Efforts?

Current state commitments, as reaffirmed at COP30, do not align with the goal of limiting warming to +1.5°C. The projections from En-ROADS’ “Business as Usual” scenario (+3.3°C by 2100) are consistent with this insufficient trajectory. Achieving the Paris Agreement goals therefore requires a major surge in ambition.

Our comprehensive scenario (+2.3°C) demonstrates that combined and systemic action – acting simultaneously on demand, energy supply, natural sinks, and short-lived gases – makes it possible to get closer to the +2°C target. However, to aim for +1.5°C, even greater efforts would be required:

- A higher and earlier carbon tax;
- An energy efficiency rate higher than +2.5%/yr;
- Even faster electrification of uses;
- Massive deployment of renewables coupled with technological advances in storage;
- A more ambitious reduction in deforestation than -50%/yr.

The challenge is therefore immense, but it remains technically accessible. The main difficulty now lies in the political and social implementation of these transformations, as shown by the complex negotiations of COP30. The gap between stated intentions and concrete actions must be bridged quickly to hope to contain warming at sustainable levels.

9 Limitations of the Model

Although the En-ROADS simulator is a powerful tool for understanding climate dynamics, it has several limitations that should be highlighted to correctly interpret our results.

9.1 A Techno-Centric Approach that Neglects Social Factors

The model quantifies the transition in terms of technological deployment and energy efficiency but does not simulate the social acceptability of measures. A high carbon tax or a radical dietary change may seem effective on paper, but their real feasibility depends on complex human factors: social movements, political instability, energy poverty, or inequalities in access to low-carbon solutions. Our scenarios therefore rely on implicit assumptions of acceptability that are not guaranteed in reality.

9.2 The Absence of a Budget Constraint

Unlike states that have limited budgets, En-ROADS does not manage a finite "bank account". The user can simultaneously activate massive subsidies for renewables, electrification, and reforestation without the model signaling a lack of public funds. While the simulator calculates the impact of taxes on energy prices, it does not account for the public debt needed to finance these transformations. This absence of a financial constraint can give the illusion that the transition depends solely on political will, whereas it requires a real mobilization of considerable capital.

9.3 A Global Model Without Regional Resolution

En-ROADS treats the planet as a single aggregated block, which masks geographical and geopolitical disparities. One cannot distinguish the respective efforts of China, the European Union, or developing countries, nor represent diplomatic tensions or trade agreements. A measure applied in the model is assumed to be applied everywhere at the same time, obscuring differences in wealth, natural resources, and institutional capacities between nations.

9.4 Limitations Related to Land Use

Internally, En-ROADS uses regionalized data to initialize carbon fluxes from forests and soils. However, from the user's perspective, the model is presented in an aggregated manner at the global scale, which poses several limitations:

- The user does not see where new forests are located. However, the albedo effect varies with latitude: planting trees in snowy areas can warm the planet instead of cooling it.
- Conflicts of use with agricultural land (food security) are not apparent.
- Biodiversity and ecosystem preservation are not explicitly modeled.

Thus, although the underlying calculations rely on regionalization, their aggregation in the interface masks part of the geographical and ecological complexity related to land use.

9.5 Non-Linearity of Policies and the Long Timeframe

Finally, the model integrates infrastructure inertia (buildings, power plants, vehicles) but does not account for political and social delays. Implementing standards, subsidies, or taxes takes time, encounters resistance, and requires compromises. Our scenarios assume a linear and continuous application of measures, whereas in reality, climate policies experience slowdowns, reversals, or adjustments that considerably lengthen transition timelines.

These limitations do not undermine the model's usefulness but remind us that its results should be interpreted as trends and orders of magnitude, not as exact predictions. They provide a solid basis for reflection, to be compared with economic, social, and geopolitical realities.

10 Conclusion

This project allowed us to demonstrate, via the En-ROADS simulator, that no single climate lever is sufficient to meet the Paris Agreement. Carbon tax, energy efficiency, electrification, reforestation, or methane reduction: each produces effects, but their impacts remain limited alone.

Our comprehensive scenario, combining all these levers in a coordinated manner, reaches $+2.3^{\circ}\text{C}$ by 2100 compared to $+3.3^{\circ}\text{C}$ without action. This result proves that synergies between sub-systems are the key to an effective transition.

COP30 in November 2025 showed that technical solutions exist but that the bottleneck is political and social. The model's limitations (absence of budget constraints, unmodeled social factors, global scale) remind us that modeling informs without predicting.

This work confirms that the transition is possible, but that it requires political will, massive investments, and collective mobilization.